

DNN Solved Workbook

Repeated PYQ Topics

Perceptron, Logistic Regression, Softmax, CNN output shapes, Backpropagation, Transfer Learning, RNN/GRU, Attention.

Perceptron Solution

$z = w \cdot x + b$. Predict using step activation. Update: $w_{\text{new}} = w + \eta(y - \hat{y})x$. Practice AND/OR gates.

Gradient Descent

Compute gradient, substitute values, update weights using $w = w - \eta \cdot \text{grad}$.

Logistic Regression

1. Compute z . 2. Apply sigmoid. 3. Compute $\text{gradient} = (\hat{y} - y)x$. 4. Update weights.

Softmax Example

Compute $\exp(z_i)$, normalize by sum, compute $\text{CCE} = -\log(\text{probability of true class})$.

CNN Output

$\text{Output} = (N - F + 2P) / S + 1$. Example 8x8 with 3x3 gives 6x6.

RNN/GRU

GRU preserves memory using update gate z_t . Helps vanishing gradients.

Attention

$\text{score} = QK^T / \sqrt{d_k}$, softmax, weighted sum of V .